

$$F_{n+1} = F_n + F_{n-1}$$

$$\begin{bmatrix} F_{n+1} \\ F_n \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} F_n \\ F_{n-1} \end{bmatrix}$$

$$B = A \cdot X$$

Initial terms $\Rightarrow \begin{bmatrix} F_n \\ F_{n-1} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

eigenvalues:-

$$\begin{vmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{vmatrix} = |A - I\lambda|$$

$$\begin{vmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{vmatrix} = 0$$

$$-\lambda(1-\lambda) - 1 = 0$$

$$-\lambda + \lambda^2 - 1 = 0$$

$$\lambda^2 - \lambda - 1 = 0$$

$$\lambda = \frac{1 \pm \sqrt{1+4}}{2}$$

$$\lambda = \frac{1 + \sqrt{5}}{2} \text{ or } \lambda = \frac{1 - \sqrt{5}}{2}$$

eigenvectors:-
for $\lambda = \frac{1+\sqrt{5}}{2}$

$$\begin{bmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} 1-\lambda & 1 & | & 0 \\ 1 & -\lambda & | & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1/1-\lambda & | & 0 \\ 1 & -\lambda & | & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 1/1-\lambda & | & 0 \\ 0 & -\lambda & | & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -\lambda & | & 0 \\ 0 & 0 & | & 0 \end{bmatrix}$$

$$X = \begin{bmatrix} \frac{1+\sqrt{5}}{2} \\ 1 \end{bmatrix} x_1$$

for $\lambda = \frac{1-\sqrt{5}}{2}$

$$\begin{bmatrix} \frac{1+\sqrt{5}}{2} & 1 \\ 1 & -\frac{1+\sqrt{5}}{2} \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & -\frac{1+\sqrt{5}}{2} \\ 1 & -\frac{1+\sqrt{5}}{2} \end{bmatrix}$$

$$x = \begin{bmatrix} \frac{1-\sqrt{5}}{2} \\ 1 \end{bmatrix} x_0$$

$$A = P D P^{-1}$$

$$A = \begin{bmatrix} \frac{1+\sqrt{5}}{2} & \frac{1-\sqrt{5}}{2} \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \frac{1+\sqrt{5}}{2} & 0 \\ 0 & \frac{1-\sqrt{5}}{2} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{5}} & \frac{-1+\sqrt{5}}{2\sqrt{5}} \\ -\frac{1}{\sqrt{5}} & \frac{1+\sqrt{5}}{2\sqrt{5}} \end{bmatrix}$$

$$A^{n-1} = \begin{bmatrix} \phi^n & \psi^n \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \phi^{n-1} & 0 \\ 0 & \psi^{n-1} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{5}} & \frac{-\psi}{\sqrt{5}} \\ -\frac{1}{\sqrt{5}} & \frac{\phi}{\sqrt{5}} \end{bmatrix}$$

$$= \begin{bmatrix} \phi^n & \psi^n \\ \phi^{n-1} & \psi^{n-1} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{5}} & \frac{-\psi}{\sqrt{5}} \\ -\frac{1}{\sqrt{5}} & \frac{\phi}{\sqrt{5}} \end{bmatrix}$$

$$A^{n-1} = \begin{bmatrix} \phi^n/\sqrt{5} - \psi^n/\sqrt{5} & -\phi^n\psi/\sqrt{5} + \psi^n\phi/\sqrt{5} \\ \phi^{n-1}/\sqrt{5} - \psi^{n-1}/\sqrt{5} & -\phi^{n-1}\psi/\sqrt{5} + \psi^{n-1}\phi/\sqrt{5} \end{bmatrix}$$

$$\begin{bmatrix} F_{n+1} \\ F_n \end{bmatrix} = A^{n-1} x \Rightarrow x = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ } \because \text{initial term}$$

$$\begin{bmatrix} F_{n+1} \\ F_n \end{bmatrix} = \begin{bmatrix} \phi^n/\sqrt{5} - \psi^n/\sqrt{5} \\ \phi^{n-1}/\sqrt{5} - \psi^{n-1}/\sqrt{5} \end{bmatrix}$$

$$F_{n+1} = \frac{\phi^n - \psi^n}{\sqrt{5}}$$